

# **Lecture**

# **Pattern Matching and Machine Learning**

# **for Audio Signal Processing**

# **MA-INF 2212**

§0 Organizational Issues

**Prof. Dr. Frank Kurth**

Permission to record the lecture or parts thereof is *not* granted!

# Lecture & Exercise Times and Locations

Lecture: Thursdays, **16:15 s.t. (i.e., sharp!)** – 17:45 h

- The lecture is **regularly held in presence** (HSZ, HS 3)
- In **exceptional cases only** (to be announced separately), the lecture will **be held in Zoom** under

<https://uni-bonn.zoom.us/j/92984663754?pwd=QVIPSk9oNktWT1FNQVJJM0hVR3c2Zz09>

Meeting-ID: 929 8466 3754, Password: 230420

Exercises: Thursdays **18:00** – 19:00 [**First exercises: 30.04.2026**]

- The exercises will be also held in presence (HSZ, HS 3)
- In **exceptional cases only** the exercises will **be held in Zoom** under

<https://uni-bonn.zoom.us/j/94424062258?pwd=SUhHWFB0a0k3MVJ4RkRISmE4S2JUUT09>

Meeting-ID: 944 2406 2258, Password: 159168

**Register** for the lecture/exam **in BASIS as usual!**

**Registration** will be open **until May 05th, 2026, 23:55 h**

- Enter the lecture / Zoom **few minutes before the lecture starts**, allowing some time for setting up your own audio equipment (!)
- Everything should work fine as long as **we all keep up some discipline** 😊:
  - Video transmission from participants will be disabled by default
  - Audio transmission from participants will be muted by default
  - **You can unmute your audio manually – please only do so for questions!**
  - Lecture will stop from time to time to allow for questions (QA)
  - Lecture will be (partially) recorded – recording will stop during QA
  - Zoom gives a clear visual indication while the recording is going on, and it automatically plays an audio message when the recording is started or paused.
  - **Please do not transmit audio while the recording is running. In doing so, you automatically consent to having your voice in the recording!**

# Exercises & Exercise Sheets

- Participation in the exercises strongly encouraged (1x / week).
- Each week, one exercise sheet has to be solved:
  - exercise sheets distributed by us each week (via WWW),
  - solutions to be handed in via **eCampus**. (Due date is stated on the exercise sheet),
  - will be corrected, scored and returned via email.

Exercise sheets: Only solutions submitted (a) via eCampus und (b) by Friday the latest (see schedule on next page!) are accepted (no exceptions made!)

- You have to work in groups ( $2 \leq N \leq 4$  students)
  - Each group hands in a „group solution“ of the exercise sheets.
  - Note however: *Each group member has to be able to demonstrate all of the groups' solutions!*

— Solutions that are obviously copied from other groups will be rejected!

- Exercises coordinated and given by: Mrs. Alessia Cornaggia-Urrigshardt

# Schedule: Lectures & Exercise Sheets

|       |              |                        |     |     |     |                    |     |     |     |     |     |
|-------|--------------|------------------------|-----|-----|-----|--------------------|-----|-----|-----|-----|-----|
|       | Week         |                        |     | 15  |     | 16                 |     | 17  |     | 18  |     |
|       |              | Thu                    | Fri | Thu | Fri | Thu                | Fri | Thu | Fri | Thu | Fri |
|       | <b>April</b> |                        |     |     |     | 16.                | 17. | 23. | 24. | 30. |     |
| Sheet | out          |                        |     |     |     | 1                  |     | 2   |     | 3   |     |
|       | in           |                        |     |     |     |                    | -   |     | 1   |     |     |
|       | scored       |                        |     |     |     | -                  |     | -   |     | 1   |     |
|       | Week         |                        |     | 19  |     | 20                 |     | 21  |     | 22  |     |
|       | <b>May</b>   |                        | 1.  | 07. | 08. | 14.                | 15. | 21. | 22. | 28. | 29. |
| Sheet | out          |                        |     | 4   |     |                    |     | 5   |     |     |     |
|       | in           |                        | 2   |     | 3   |                    | 4   |     |     |     |     |
|       | scored       |                        |     | 2   |     |                    |     | 3+4 |     |     |     |
|       | Week         |                        |     | 23  |     | 24                 |     | 25  |     | 26  |     |
|       | <b>June</b>  |                        |     | 04. | 05. | 11.<br>L<br>08.06. | 12. | 18. | 19. | 25. | 26. |
| Sheet | out          |                        |     |     |     | 6                  |     | 7   |     | 8   |     |
|       | in           |                        |     |     | 5   |                    |     |     | 6   |     | 7   |
|       | scored       |                        |     |     |     |                    |     | 5   |     | 6   |     |
|       | Week         | 27                     |     | 28  |     | 29                 |     | 30  |     |     |     |
|       | <b>July</b>  | 02.<br>L + E<br>29.06. | 03. | 09. | 10. | 16.                | 17. | 23. | 24. |     |     |
| Sheet | out          | 9                      |     | 10* |     | -                  |     | -   |     |     |     |
|       | in           |                        | 8   |     | 9   |                    | 10* |     | -   |     |     |
|       | scored       | 7                      |     | 8   |     | 9                  |     | 10* |     |     |     |

**#Exercise sheets: 10**

(10\*: Preparation for exam)

**No lecture/exercises:**

14.05., 28.05., 04.06. 

(25.-29.05. excursion week)

**Lectures & Exercises (L+E)  
held in Zoom:**

08.06. (no L+E on 11.06.)

Lecture: 16 – 18 h,

No exercises

29.06. (no L+E on 02.07.)

Lecture: 16 – 18 h,

Exercises: 18 – 19 h

**Final exam:**

18.08.2026, 09:00-11:00 h

**Resit exam:**

04.09.2026, 10:00-12:00 h

You will be **automatically registered** for the **eCampus** course when registering in BASIS.

**In eCampus** you will find

- Weekly exercise sheets
- Notes on important dates (exams,...)
- Additional literature as necessary
- Submission site for all exercises (mind the deadlines!)
- Handouts of the slides shown in the lecture (**note however, that additional material is provided during the lecture!**)
- Log in using your **Uni-ID** as user name
- Link to the eCampus course (available also from BASIS):

[https://ecampus.uni-bonn.de/ilias.php?baseClass=ilrepositorygui&ref\\_id=4053118](https://ecampus.uni-bonn.de/ilias.php?baseClass=ilrepositorygui&ref_id=4053118)

# Obtaining the 6 credit points (CPs)...

## Final exam:

Tue., 18.08.26, 9:00-11:00 h [time/location to be confirmed], (90 minutes of writing time)

Exam currently being scheduled in presence (= physically at U Bonn)

Criterion to be admitted to final exam:

Obtaining  $\geq 50$  % of all possible points/scores in the 10 exercise sheets.

- It is beneficial to actively participate in the exercises!
- If you fail the final exam, a resit exam will be offered on

04.09.2026, 10:00 – 12:00 h [time/location tbc], (90 minutes duration)

- In general: similar amount of points for all sheets
- Total of scores (T) = Sum of possible scores for sheets 1-9
- **Bonus** points for last exercise **sheet 10** (exam preparation)
- Max. available points (M) = Sum of obtained scores for sheets 1-10
  
- Criterion for exam admission:

$$M / T \geq 0.5$$



Lecturer for Computer Science at CS dept. of U Bonn

Fields of interest:

- Audio Signal Processing
- Multimedia Information Retrieval
- Efficient Algorithms
- Digital Audio and Music Libraries

Member of AG Multimedia Signal Processing (Prof. Clausen)  
1997-2006

Current position: Group Head at Fraunhofer FKIE

Contact:

Prof. Dr. Frank Kurth

Fraunhofer FKIE

Fraunhoferstr. 20

53343 Wachtberg

Tel. 0228 – 9435868

Email: [frank.kurth@fkie.fraunhofer.de](mailto:frank.kurth@fkie.fraunhofer.de)

FKIE-Office Bonn (Thursdays)

Zanderstr. 5

53177 Bonn

0228 50212-460

# Staff for this lecture

---

- Alessia Cornaggia-Urrigshardt, M.Sc. (Exercises)  
Fraunhofer FKIE  
Phone: 0228 9435-736  
Email:  
[alessia.cornaggia-urrigshardt@t-online.de](mailto:alessia.cornaggia-urrigshardt@t-online.de)



- Follow-up of  
*Foundations of Audio* Signal Processing (MA-INF 2113)  
(Winter term 2025/26)
- Builds upon the audio part of that lecture.
- More in-depth treatment of signal processing background.
- Applications from
  - Audio Signal Processing
  - Audio and Music Information Retrieval

# Seminar Audio MA-INF 2208

---

- Offered next winter semester 2026/27
- Various topics in Audio Signal processing
- Talks will be scheduled mid Jan/early Feb 2027
- In case of interest, please contact us via email any time during this summer term!

# And now...

---

... lets go for the contents!